

The Hydrogen Atom, Part II

Radial Properties, the Virial Theorem, and Angular Solutions

Quantum Mechanics Lecture Notes

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Contents

1	Radial solutions and their properties	1
1.1	Behavior at the origin	2
1.2	Behavior at large radius	2
1.3	Radial and angular nodes	2
1.4	Radial probability distribution	2
1.5	Expectation values	3
2	The quantum virial theorem	3
2.1	Why the dilation operator generates scale transformations	4
3	Angular solutions	5
3.1	Separation of the angular equation	6
3.2	Spherical harmonics	6
4	Spherical harmonics and angular momentum	7
5	Orbital shapes and degeneracies	7
5.1	The p orbitals as two different bases	7
5.2	Brief extension to the d orbitals	9

1 Radial solutions and their properties

For a hydrogenlike atom of nuclear charge Z , separation of the time-independent Schrödinger equation gives

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r)Y_{\ell}^m(\theta, \phi). \quad (1)$$

The normalized radial wavefunctions can be written as

$$R_{n\ell}(r) = N_{n\ell} \left(\frac{2Zr}{na_0} \right)^{\ell} \exp \left(-\frac{Zr}{na_0} \right) L_{n-\ell-1}^{2\ell+1} \left(\frac{2Zr}{na_0} \right), \quad (2)$$

where L_q^{α} is an associated Laguerre polynomial and

$$N_{n\ell} = \left(\frac{2Z}{na_0} \right)^{3/2} \sqrt{\frac{(n-\ell-1)!}{2n(n+\ell)!}}. \quad (3)$$

For a finite nuclear mass, the Bohr radius is defined using the electron–nucleus reduced mass μ :

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{\mu e^2}. \quad (4)$$

The allowed quantum numbers are

$$n = 1, 2, 3, \dots, \quad \ell = 0, 1, \dots, n-1, \quad m = -\ell, -\ell+1, \dots, \ell. \quad (5)$$

1.1 Behavior at the origin

Near the origin, the associated Laguerre polynomial approaches a constant, so

$$R_{n\ell}(r) \sim r^\ell \quad (r \rightarrow 0). \quad (6)$$

Consequently, only the $\ell = 0$ states (the s states) can have nonzero wavefunction amplitude at the origin. All states with $\ell > 0$ vanish there.

1.2 Behavior at large radius

Every bound-state radial function is a polynomial in r multiplying the exponential factor

$$R_{n\ell}(r) \sim \text{poly}(r) \exp\left(-\frac{Zr}{na_0}\right) \quad (r \rightarrow \infty). \quad (7)$$

Thus all bound states decay exponentially. The characteristic radial scale is of order $n^2 a_0 / Z$: excited states extend farther from the nucleus as n increases.

1.3 Radial and angular nodes

The associated Laguerre polynomial in Eq. (2) has degree $n - \ell - 1$. Therefore,

$$N_{\text{radial}} = n - \ell - 1, \quad (8)$$

$$N_{\text{angular}} = \ell, \quad (9)$$

$$N_{\text{total}} = n - 1. \quad (10)$$

Radial nodes occur at finite positive values of r for which $R_{n\ell}(r) = 0$. Angular nodes are nodal surfaces of Y_ℓ^m .

1.4 Radial probability distribution

The probability of finding the electron in a volume element is

$$dP = |\psi(r, \theta, \phi)|^2 r^2 dr d\Omega, \quad d\Omega = \sin\theta d\theta d\phi. \quad (11)$$

Because the spherical harmonics are normalized, $\int |Y_\ell^m|^2 d\Omega = 1$, integration over solid angle gives

$$dP = P_{n\ell}(r) dr, \quad \boxed{P_{n\ell}(r) = r^2 |R_{n\ell}(r)|^2}. \quad (12)$$

The radial probability distribution is therefore not simply $|R|^2$; the spherical volume factor r^2 is essential.

For the ground state,

$$R_{10}(r) = 2 \left(\frac{Z}{a_0} \right)^{3/2} \exp \left(-\frac{Zr}{a_0} \right), \quad (13)$$

and hence

$$P_{10}(r) = 4 \left(\frac{Z}{a_0} \right)^3 r^2 \exp \left(-\frac{2Zr}{a_0} \right). \quad (14)$$

Although $|R_{10}(r)|^2$ is largest at the origin, $P_{10}(0) = 0$. The radial probability is maximal at

$$\boxed{r_{\text{mp}} = \frac{a_0}{Z}}, \quad (15)$$

the Bohr radius for hydrogen ($Z = 1$).

1.5 Expectation values

A useful general result is

$$\boxed{\langle r \rangle_{n\ell} = \frac{a_0}{2Z} [3n^2 - \ell(\ell + 1)]}. \quad (16)$$

For the ground state,

$$r_{\text{mp}} = \frac{a_0}{Z}, \quad \langle r \rangle_{10} = \frac{3a_0}{2Z}. \quad (17)$$

The most probable radius and the mean radius are different quantities. We will also derive below that

$$\boxed{\left\langle \frac{1}{r} \right\rangle_{n\ell} = \frac{Z}{a_0 n^2}}. \quad (18)$$

In general, $\langle 1/r \rangle \neq 1/\langle r \rangle$.

2 The quantum virial theorem

Consider

$$H = T + V = \frac{\mathbf{p}^2}{2\mu} + V(\mathbf{r}) \quad (19)$$

and introduce the Hermitian dilation operator

$$\boxed{G = \frac{1}{2} (\mathbf{r} \cdot \mathbf{p} + \mathbf{p} \cdot \mathbf{r})}. \quad (20)$$

For a stationary state $H|n\rangle = E_n|n\rangle$,

$$\langle [H, G] \rangle_n = \langle n | HG | n \rangle - \langle n | GH | n \rangle = 0. \quad (21)$$

Using $[r_i, p_j] = i\hbar\delta_{ij}$ gives

$$[T, G] = -2i\hbar T, \quad (22)$$

$$[V, G] = i\hbar \mathbf{r} \cdot \nabla V. \quad (23)$$

Therefore

$$[H, G] = i\hbar (\mathbf{r} \cdot \nabla V - 2T), \quad (24)$$

and its expectation value yields the quantum virial theorem:

$$\boxed{2 \langle T \rangle = \langle \mathbf{r} \cdot \nabla V \rangle}. \quad (25)$$

For a central potential $V = V(r)$,

$$\mathbf{r} \cdot \nabla V = r \frac{dV}{dr}. \quad (26)$$

If the potential is homogeneous of degree k ,

$$V(a\mathbf{r}) = a^k V(\mathbf{r}), \quad (27)$$

Euler's theorem gives $\mathbf{r} \cdot \nabla V = kV$. Equation (25) then becomes

$$\boxed{2 \langle T \rangle = k \langle V \rangle}. \quad (28)$$

For the Coulomb potential, $V(r) \propto r^{-1}$, so $k = -1$ and

$$\boxed{2 \langle T \rangle = - \langle V \rangle}. \quad (29)$$

It follows that

$$E = \langle T \rangle + \langle V \rangle, \quad \boxed{\langle T \rangle = -E}, \quad \boxed{\langle V \rangle = 2E}. \quad (30)$$

For the remainder of this subsection, set $4\pi\epsilon_0 = 1$. Then

$$V(r) = -\frac{Ze^2}{r}, \quad E_n = -\frac{\mu Z^2 e^4}{2\hbar^2 n^2} = -\frac{e^2 Z^2}{2a_0 n^2}, \quad a_0 = \frac{\hbar^2}{\mu e^2}. \quad (31)$$

Using $\langle V \rangle = 2E_n$,

$$-Ze^2 \left\langle \frac{1}{r} \right\rangle = -\frac{\mu Z^2 e^4}{\hbar^2 n^2}, \quad (32)$$

which immediately recovers Eq. (18):

$$\boxed{\left\langle \frac{1}{r} \right\rangle = \frac{\mu Z e^2}{\hbar^2 n^2} = \frac{Z}{a_0 n^2}}. \quad (33)$$

2.1 Why the dilation operator generates scale transformations

The role of G is deeper than its use in the virial theorem. Define the unitary operator

$$U(\lambda) = \exp\left(-\frac{i\lambda G}{\hbar}\right). \quad (34)$$

The commutators

$$[G, r_i] = -i\hbar r_i, \quad [G, p_i] = +i\hbar p_i \quad (35)$$

imply, for $r_i(\lambda) = U^\dagger(\lambda) r_i U(\lambda)$ and similarly for $p_i(\lambda)$,

$$\frac{dr_i(\lambda)}{d\lambda} = r_i(\lambda), \quad (36)$$

$$\frac{dp_i(\lambda)}{d\lambda} = -p_i(\lambda). \quad (37)$$

Thus

$$\boxed{\mathbf{r}(\lambda) = e^\lambda \mathbf{r}, \quad \mathbf{p}(\lambda) = e^{-\lambda} \mathbf{p}}. \quad (38)$$

The reciprocal scaling preserves the canonical commutator $[r_i, p_j] = i\hbar \delta_{ij}$.

In the coordinate representation, $\mathbf{p} = -i\hbar \nabla$. In three dimensions,

$$\mathbf{r} \cdot \mathbf{p} = -i\hbar \mathbf{r} \cdot \nabla, \quad (39)$$

$$\mathbf{p} \cdot \mathbf{r} = -i\hbar (\mathbf{r} \cdot \nabla + 3), \quad (40)$$

so

$$\boxed{G = -i\hbar \left(\mathbf{r} \cdot \nabla + \frac{3}{2} \right)}. \quad (41)$$

Acting on a wavefunction,

$$\boxed{\psi_\lambda(\mathbf{r}) = (U(\lambda)\psi)(\mathbf{r}) = e^{-3\lambda/2} \psi(e^{-\lambda} \mathbf{r})}. \quad (42)$$

The factor $e^{-3\lambda/2}$ ensures that normalization is preserved.

Under a dilation, the kinetic and homogeneous potential terms scale as

$$T \longrightarrow e^{-2\lambda} T, \quad V \longrightarrow e^{k\lambda} V. \quad (43)$$

Thus the energy expectation in the scaled state is

$$E(\lambda) = e^{-2\lambda} \langle T \rangle + e^{k\lambda} \langle V \rangle. \quad (44)$$

An energy eigenstate is stationary under an infinitesimal change of spatial scale:

$$\left. \frac{dE}{d\lambda} \right|_{\lambda=0} = -2 \langle T \rangle + k \langle V \rangle = 0. \quad (45)$$

This is again the virial theorem. Equivalently, $[H, G]$ measures the response of the Hamiltonian to an infinitesimal spatial dilation.

3 Angular solutions

Write the separated wavefunction as

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi). \quad (46)$$

The angular eigenvalue equation is

$$\Lambda^2 Y = \lambda Y, \quad (47)$$

where the dimensionless angular operator is

$$\boxed{\Lambda^2 = - \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]}. \quad (48)$$

3.1 Separation of the angular equation

Set

$$Y(\theta, \phi) = \Theta(\theta)\Phi(\phi). \quad (49)$$

Substitution gives

$$-\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \frac{1}{\Phi \sin^2 \theta} \frac{d^2 \Phi}{d\phi^2} = \lambda. \quad (50)$$

The ϕ -dependent term must be constant. Writing that constant as m^2 gives

$$\frac{d^2 \Phi}{d\phi^2} + m^2 \Phi = 0, \quad \Phi(\phi) = A e^{im\phi} + B e^{-im\phi}. \quad (51)$$

For an L_z eigenstate, choose $\Phi_m \propto e^{im\phi}$. Single-valuedness requires

$$\Phi(\phi + 2\pi) = \Phi(\phi) \implies e^{2\pi im} = 1, \quad (52)$$

and therefore

$$\boxed{m = 0, \pm 1, \pm 2, \dots}. \quad (53)$$

The polar equation is

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \left(\lambda - \frac{m^2}{\sin^2 \theta} \right) \Theta = 0. \quad (54)$$

With $x = \cos \theta$ and $d/d\theta = -\sin \theta \, d/dx$, this becomes

$$\boxed{(1-x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \left(\lambda - \frac{m^2}{1-x^2} \right) \Theta = 0}. \quad (55)$$

This is the associated Legendre equation. Regularity at the poles $x = \pm 1$ restricts the separation constant and the quantum numbers to

$$\boxed{\lambda = \ell(\ell+1)}, \quad \ell = 0, 1, 2, \dots, \quad \boxed{|m| \leq \ell}. \quad (56)$$

Thus the quantization of ℓ and m follows from the boundary conditions: single-valuedness in ϕ and regularity at the poles.

3.2 Spherical harmonics

Adopt the Condon–Shortley convention

$$P_\ell^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x) \quad (m \geq 0). \quad (57)$$

The normalized spherical harmonics are

$$\boxed{Y_\ell^m(\theta, \phi) = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\phi}} \quad (m \geq 0), \quad (58)$$

with negative- m functions defined by

$$Y_\ell^{-m} = (-1)^m (Y_\ell^m)^*. \quad (59)$$

They obey

$$\int Y_\ell^{m*}(\theta, \phi) Y_{\ell'}^{m'}(\theta, \phi) \, d\Omega = \delta_{\ell\ell'} \delta_{mm'}. \quad (60)$$

4 Spherical harmonics and angular momentum

The angular operator is the dimensionless form of orbital angular momentum:

$$\boxed{L^2 = \hbar^2 \Lambda^2}, \quad \boxed{L_z = -i\hbar \frac{\partial}{\partial \phi}}. \quad (61)$$

Therefore,

$$L^2 Y_\ell^m = \hbar^2 \ell(\ell+1) Y_\ell^m, \quad (62)$$

$$L_z Y_\ell^m = \hbar m Y_\ell^m. \quad (63)$$

For a central potential,

$$[H, L^2] = 0, \quad [H, L_z] = 0, \quad (64)$$

so energy eigenstates may also be chosen as simultaneous eigenstates of L^2 and one angular-momentum component, conventionally L_z . We cannot choose all three components simultaneously because

$$[L_x, L_y] = i\hbar L_z, \quad [L_y, L_z] = i\hbar L_x, \quad [L_z, L_x] = i\hbar L_y. \quad (65)$$

5 Orbital shapes and degeneracies

For the pure Coulomb problem, the bound-state energy depends only on n :

$$E_n = -\frac{\mu Z^2 e^4}{2(4\pi\epsilon_0)^2 \hbar^2 n^2}. \quad (66)$$

In the absence of spin, each shell therefore has degeneracy

$$\sum_{\ell=0}^{n-1} (2\ell+1) = n^2. \quad (67)$$

External fields, relativistic corrections, and other perturbations can lift this degeneracy. For $n=2$, the four spatial orbitals are one $2s$ orbital and three $2p$ orbitals.

The orbital letter labels describe the angular quantum number and the associated shape:

ℓ	label	qualitative angular shape
0	s	spherical; one angular state
1	p	two-lobed (dumbbell); three angular states
2	d	four-lobed or d_{z^2} form; five angular states

The surfaces commonly drawn in chemistry are angular-density or isosurface representations; the signs placed on lobes indicate the phase of the wavefunction, not electric charge.

5.1 The p orbitals as two different bases

For $\ell=1$,

$$Y_1^0(\theta, \phi) = \sqrt{\frac{3}{4\pi}} \cos \theta = \sqrt{\frac{3}{4\pi}} \frac{z}{r}, \quad (68)$$

$$Y_1^{+1}(\theta, \phi) = -\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi}, \quad (69)$$

$$Y_1^{-1}(\theta, \phi) = +\sqrt{\frac{3}{8\pi}} \sin \theta e^{-i\phi}. \quad (70)$$

Thus Y_1^0 is the angular part of the familiar p_z orbital:

$$Y_{p_z} = Y_1^0 = \sqrt{\frac{3}{4\pi}} \frac{z}{r}. \quad (71)$$

The $m = \pm 1$ functions are complex eigenfunctions of L_z . The familiar real p_x and p_y orbitals are linear combinations of these degenerate states:

$$Y_{p_x} = \frac{1}{\sqrt{2}} (Y_1^{-1} - Y_1^{+1}) = \sqrt{\frac{3}{4\pi}} \sin \theta \cos \phi = \sqrt{\frac{3}{4\pi}} \frac{x}{r}, \quad (72)$$

$$Y_{p_y} = \frac{i}{\sqrt{2}} (Y_1^{-1} + Y_1^{+1}) = \sqrt{\frac{3}{4\pi}} \sin \theta \sin \phi = \sqrt{\frac{3}{4\pi}} \frac{y}{r}. \quad (73)$$

The complex basis

$$\{Y_1^{+1}, Y_1^0, Y_1^{-1}\} \quad (74)$$

diagonalizes L_z , while the real Cartesian basis

$$\{Y_{p_x}, Y_{p_y}, Y_{p_z}\} \quad (75)$$

makes spatial orientation transparent. These are two bases for the same three-dimensional $\ell = 1$ subspace; they do not represent six different states. In particular,

$$L_z Y_{p_x} = i\hbar Y_{p_y}, \quad L_z Y_{p_y} = -i\hbar Y_{p_x}, \quad L_z Y_{p_z} = 0, \quad (76)$$

so p_x and p_y are not eigenstates of L_z .

5.2 Brief extension to the d orbitals

The $\ell = 2$ subspace is five-dimensional. Its complex basis has $m = -2, -1, 0, 1, 2$, while a real Cartesian basis is

$$\{d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}\}. \quad (77)$$

Their angular dependence is proportional to

$$\frac{xy}{r^2}, \quad \frac{yz}{r^2}, \quad \frac{xz}{r^2}, \quad \frac{x^2 - y^2}{r^2}, \quad \frac{3z^2 - r^2}{r^2}, \quad (78)$$

respectively. With the same phase convention used above, normalized real combinations include

$$Y_{d_{z^2}} = Y_2^0 = \sqrt{\frac{5}{16\pi}} \frac{3z^2 - r^2}{r^2}, \quad (79)$$

$$Y_{d_{x^2-y^2}} = \frac{Y_2^{-2} + Y_2^{+2}}{\sqrt{2}} = \sqrt{\frac{15}{16\pi}} \frac{x^2 - y^2}{r^2}, \quad (80)$$

$$Y_{d_{xy}} = \frac{i}{\sqrt{2}} (Y_2^{-2} - Y_2^{+2}) = \sqrt{\frac{15}{4\pi}} \frac{xy}{r^2}, \quad (81)$$

$$Y_{d_{xz}} = \frac{Y_2^{-1} - Y_2^{+1}}{\sqrt{2}} = \sqrt{\frac{15}{4\pi}} \frac{xz}{r^2}, \quad (82)$$

$$Y_{d_{yz}} = \frac{i}{\sqrt{2}} (Y_2^{-1} + Y_2^{+1}) = \sqrt{\frac{15}{4\pi}} \frac{yz}{r^2}. \quad (83)$$

As for the p orbitals, these real orbitals are simply a change of basis within the fixed- ℓ angular-momentum subspace.